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Week 5: Image Processing

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Reading: Week 5 Lecture Notebook30 min

Ungraded Plug-in: 5.1 Overview of Image Processing14 min

Ungraded Plug-in: 5.2 Pixel Processing3 min

Practice Quiz: 5.2 Pixel Processing Self-check QuizStarted

Reading: 5.3 Video10 min

Ungraded Plug-in: 5.3 LSI and Convolution22 min

Practice Quiz: 5.3 LSI and Convolution Self-check QuizStarted

Reading: 5.4 Video10 min

Ungraded Plug-in: 5.4 Linear Image Filters16 min

Practice Quiz: 5.4 Linear Image Filters Self-check Quiz15 min

Ungraded Plug-in: 5.5 Non-Linear Image Filters16 min

Practice Quiz: 5.5 Non-Linear Image Filters Self-check Quiz15 min

Reading: 5.6 Video10 min

Ungraded Plug-in: 5.6 Template Matching by Correlation7 min

Practice Quiz: 5.6 Template Matching by Correlation Self-check Quiz10 min

Discussion Prompt: Week 5: Analyzing Convolution10 min

Discussion Prompt: Week 5: Questions and Feedback

Week 5 Quiz

🎉 Congratulations! You passed!

Grade received 100%. To pass 50% or higher

Go to next item

5.3 LSI and Convolution Self-check Quiz

1. Two systems with impulse responses $h_1(t)$ and $h_2(t)$ are connected in a cascade. The overall impulse response of the cascaded system is given by. 1 / 1 point

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Product of $h_1(t)$ and $h_2(t)$

Sum of $h_1(t)$ and $h_2(t)$

Convolution of $h_1(t)$ and $h_2(t)$

Convolution of product of $h_1(t)$ and $h_2(t)$ with input signal

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Help me practice

Let's chat

👍 **Correct**
Convolution of $h_1(t)$ and $h_2(t)$ **is the right answer**

Try again

2. Which of the following is correct regarding impulse signal? 1 / 1 point

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$x[n]\delta[n] = \delta[n]$

$x[n]\delta[n] = x[0]\delta[n]$

$x[n]\delta[n] = x[n]$

$x[n]\delta[n] = x[0]$

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Correct

When the input $x[n]$ is multiplied with an impulse signal, the result will be an impulse signal with magnitude of $x[n]$ at that time.

3. Which of the following is correct regarding impulse signal? 1 / 1 point

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$x[n] * \delta[n] = \delta[n]$

$x[n] * \delta[n] = x[0]\delta[n]$

$x[n] * \delta[n] = x[n]$

$x[n] * \delta[n] = x[0]$

👎

👍

Correct

When the input $x[n]$ is convolved with an impulse signal, the result will be the original signal.

https://www.coursera.org/learn/cameraandimaging/quiz/AK8bU5-3-lsi-and-convolution-self-check-quiz?attemptRedirectToCover=true

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